

TE_2.3: Unified Picture

Universal Generative Principle (UGP)

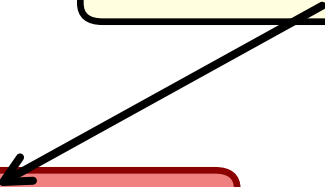


Generative Theory of Everything (GTE)
(Discrete triple structure)



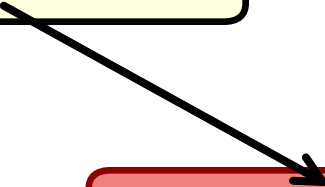
Self-Referential RG (SRRG) Flow
(Viability functional $F[S]$)

*Gauge group:
 $SU(3) \times SU(2) \times U(1)$
3 generations
Quarter-Lock*



Standard Model

*Binding energies
Magic numbers
Shell structure
Island of stability*



Nuclear Physics

- TS1: 97% attraction
- TS3: Gauge running
 - TS9: c-function
- UGP_lab: $\theta_W \approx \pi/12$

- TS5: 0.48 MeV MAE
- PERIODIC_TABLE_APP
- AME-2020: 0.489 MeV
- 2,457 nuclei, $R^2 = 0.9996$

Same UGP/GTE/SRRG structure uniquely determines both SM gauge couplings and nuclear binding energies